

SASA-ESM: Surface-Aware Equivariant Residue Modeling for Protein–Protein Interaction Site Prediction

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Abstract. Protein–protein interaction (PPI) site prediction requires residue-level reasoning over biochemical exposure, sequence context, and three-dimensional geometry. We present SASA-ESM, a reproducible pipeline that constructs weak residue labels from Δ solvent-accessible surface area (SASA), represents each residue with ESM-2 protein language model embeddings and local structural descriptors, and trains an E(n)-equivariant graph neural network (EGNN) for interface classification. The pipeline uses no external graph packages: SASA, Δ SASA labels, residue features, graph construction, EGNN training, optional cross-chain analysis, benchmark manifests, and prediction exports are implemented in the accompanying code. On a 500-complex training corpus (95,005 target-chain residues), the leakage-controlled ESM-2 650M plus geometry EGNN with an 8 Å residue graph cutoff reaches test F1 = 0.7745, AUROC = 0.9322, and AUPRC = 0.8421. Adding apo SASA changes F1 only slightly to 0.7787. In contrast, adding holo SASA raises F1 to 0.8948, demonstrating why holo-aware features must be reported only as leakage diagnostics when labels are defined by $\text{SASA}^{apo} - \text{SASA}^{holo}$. Strict external evaluation reaches F1 = 0.3529 on Dset_186-local and = 0.3040 on PDBtest_315-local. These results support a reproducible surface-labeled pipeline while exposing the remaining generalization gap.

Keywords: Protein–protein interaction · Solvent-accessible surface area · Protein language model · ESM-2 · Equivariant graph neural network

1 Introduction

Identifying the amino-acid residues that form a protein–protein interface is a central problem for structural bioinformatics, drug discovery, protein engineering, and functional annotation. Experimental structures provide the most direct evidence of interacting surfaces, but manually deriving residue-level labels is laborious and sensitive to the definition of contact. A practical predictor must combine at least three complementary signals: surface exposure, evolutionary sequence context, and geometric compatibility between nearby residues.

Solvent-accessible surface area (SASA) is a natural physical measure for this task. A residue that is exposed in an isolated chain and becomes buried in a

complex is likely to participate in the interface. This observation motivates the common Δ SASA label:

$$\Delta\text{SASA}_i = \text{SASA}_i^{\text{apo}} - \text{SASA}_i^{\text{holo}}, \quad y_i = \mathbb{I}[\Delta\text{SASA}_i \geq \tau].$$

However, SASA alone is not sufficient: exposed non-interface residues are frequent, and interfaces depend on residue type, local secondary geometry, and spatial organization. Recent protein language models such as ESM-2 encode rich sequence context [4], while equivariant graph neural networks can process residue coordinates without discarding the symmetries of three-dimensional space [6].

This work integrates these ingredients into SASA-ESM. The contribution is deliberately engineering-focused and reproducible: rather than assuming external preprocessing, the project implements the full data path from PDB structures to residue-level predictions. The main contributions are:

- a weakly supervised Δ SASA labeling pipeline for two-chain protein complexes;
- a leakage-controlled true ESM-2 650M residue representation combining 1,280-dimensional protein language model embeddings, backbone/HSE descriptors, hydrophobicity, and C α coordinates;
- an EGNN interface predictor, with cross-chain distance attention retained as an analysis module rather than a consistently improving core method;
- benchmark-ready prediction export on Dset_186-local and PDBtest_315-local, plus controlled feature and graph-cutoff ablations.

2 Related Work

Classical interface predictors often use geometric surface descriptors, conservation, physicochemical residue properties, or hand-designed contact features. SASA-based analysis traces back to the rolling-sphere definition of molecular surface accessibility [3], and efficient implementations such as FreeSASA made large-scale surface calculations practical [5]. Graph-based deep learning has since become a common choice for protein interface modeling because residue graphs make local neighborhoods explicit. GraphPPIS is a representative residue graph model for PPI site prediction and introduced widely used benchmark assets such as Test_315 [7].

Protein language models add another axis of information. ESM-2 learns representations from large protein sequence corpora and has been shown to encode structural and functional regularities [4]. In parallel, equivariant networks such as EGNN provide a principled way to use coordinates while respecting translation, rotation, and reflection symmetries [6]. SASA-ESM combines these lines: the label is defined by a physical surface criterion, the node representation includes language-model and local structural features, and the graph model is equivariant.

3 Method

3.1 Data Construction and Labeling

The corpus is built from two-chain protein assemblies downloaded from the RCSB Protein Data Bank [1]. Each complex is parsed into atoms, residues, chains, and $C\alpha$ residue coordinates. We retain target and partner chains with at least 30 residues to avoid degenerate small fragments.

For each target-chain residue, SASA is calculated twice. The apo value is computed on the target chain alone. The holo value is computed after placing the target chain in the presence of the partner chain. A residue is labeled as interface when $\Delta\text{SASA} \geq 2.0 \text{ \AA}^2$. This threshold is used throughout the reported experiments. Importantly, the strict protocol uses SASA only to define weak labels. Neither ΔSASA nor its holo component is used as an input feature. We separately report apo-only and holo-aware runs to quantify label-definition leakage.

3.2 Residue Representation

In the strict primary model, each residue node is represented by:

$$x_i = [e_i, s_i],$$

where $e_i \in \mathbb{R}^{1280}$ is the ESM-2 650M residue embedding and s_i contains backbone dihedral encodings ($\sin \phi, \cos \phi, \sin \psi, \cos \psi$), half sphere exposure counts, and hydrophobicity. The resulting strict true-650M feature set contains 1,287 scalar input features per residue. An apo-only ablation appends $\text{SASA}_i^{\text{apo}}$, while the explicitly labeled holo-aware diagnostic appends both apo and holo SASA. The code validates dimensionality to avoid the common failure mode of treating smaller 320-dimensional ESM features as 650M features.

3.3 Equivariant Residue Graph

For each target chain, a residue graph $G = (V, E)$ is built from $C\alpha$ coordinates. Residues i and j are connected when $\|p_i - p_j\|_2 \leq r$, where r is the distance cutoff. We evaluate $r \in \{6, 8, 10, 12\} \text{ \AA}$.

The EGNN updates node states and coordinates through messages

$$m_{ij} = \phi_e(h_i, h_j, \|p_i - p_j\|_2/10),$$

with residual feature updates and bounded coordinate weights. The implementation uses unit direction vectors and normalized distances for numerical stability. After three EGNN layers, an MLP classifier outputs an interface logit for every residue.

PDB complex $\rightarrow$ apo/holo SASA $\rightarrow$ Δ SASA labels $\rightarrow$ ESM-2 650M residue
 embeddings $\rightarrow$ ESM + structural node features $\rightarrow$ EGNN $\rightarrow$ residue interface
 probabilities

Fig. 1. SASA-ESM processing pipeline. The same manifest drives label generation, embedding extraction, multimodal dataset construction, training, and benchmark prediction export.

Table 1. Datasets used in the current package.

Dataset	Complexes	Residues	Role
Main two-chain corpus	500	95,005	Train/validation/test split
Dset_186-local	158	39,505	External benchmark prediction
PDBtest_315-local	314	65,119	External benchmark prediction

3.4 Cross-Chain Spatial Attention

The cross-chain variant augments the target-chain EGNN output with partner-chain $C\alpha$ positions. For target residue i and partner residue j , attention is computed as

$$\alpha_{ij} = \text{softmax}_j(-\|p_i - q_j\|_2/\sigma),$$

where σ is learnable. Partner coordinates are projected into hidden vectors and combined with target hidden states before classification. This lightweight mechanism lets the model access the spatial layout of the interacting chain without constructing a full bipartite atom graph.

4 Experimental Setup

4.1 Datasets

Table 1 summarizes the datasets used in this manuscript. The main corpus contains 500 two-chain complexes and 95,005 labeled target-chain residues. The Dset_186 evaluation uses the locally analyzable subset produced by the benchmark pipeline after structure availability and chain-quality filtering: 158 complexes and 39,505 residues. PDBtest_315-local uses 314 analyzable complexes and 65,119 residues after the same chain-quality rules. We use the *-local* suffix because both datasets are produced by this repository’s reproducible preprocessing protocol.

4.2 Training Protocol

Models are trained with Adam, learning rate 10^{-3} , weight decay 10^{-4} , hidden dimension 128, dropout 0.4, three EGNN layers, gradient clipping at 1.0, and early stopping based on validation F1. Complexes are split by complex identity into 70/15/15 train/validation/test partitions with seed 42. Metrics include

Table 2. Held-out test performance. Holo-aware rows are leakage diagnostics, not strict predictors.

Role	Model / features	F1	AUROC	AUPRC	Inputs
Strict baseline	apo-SASA rank	0.3317	0.5841	0.2550	apo SASA
Strict baseline	MLP / apo	0.4292	0.7000	0.3098	apo SASA
Strict baseline	MLP / ESM	0.6290	0.8949	0.7387	ESM
Strict baseline	GCN / ESM+geom.	0.5961	0.8730	0.6570	ESM+geom.
Strict primary	EGNN / ESM+geom.	0.7745	0.9322	0.8421	ESM+geom.
Strict ablation	EGNN / apo+ESM+geom.	0.7787	0.9348	0.8514	apo+ESM+geom.
Diagnostic	MLP / apo+holo	0.8724	0.9457	0.9117	apo+holo SASA
Diagnostic	EGNN / apo+holo+ESM+geom.	0.8948	0.9768	0.9497	holo-aware

Table 3. Holo-aware EGNN cutoff diagnostic. These rows are not strict predictors.

Cutoff	Acc.	Prec.	Rec.	F1	AUROC / AUPRC
6 Å	0.9574	0.9027	0.8781	0.8902	0.9750 / 0.9420
8 Å	0.9590	0.9023	0.8874	0.8948	0.9768 / 0.9497
10 Å	0.9577	0.9040	0.8777	0.8907	0.9758 / 0.9488
12 Å	0.9584	0.9188	0.8645	0.8908	0.9727 / 0.9449

accuracy, precision, recall, F1, AUROC, and AUPRC. AUPRC is reported because interface labels are class-imbalanced and precision–recall curves are more informative under skewed class priors [2].

5 Results

5.1 Leakage-Controlled Main Corpus

Table 2 reports strict baselines and leakage diagnostics. The strict ESM plus geometry EGNN improves over MLP and GCN baselines. Adding apo SASA has only a small effect. Adding holo SASA produces a much larger score increase because the label itself is computed from apo minus holo SASA; this configuration is diagnostic and is not treated as the deployable result.

5.2 Distance Cutoff Ablation

The graph cutoff controls the amount of geometric context available to each residue. Table 3 shows that 8 Å yields the best F1 and AUPRC, while larger cutoffs do not improve the held-out test set. This suggests that the most useful information is local but not extremely short-range; a 6 Å graph may miss relevant neighbors, whereas 10–12 Å adds edges with weaker interface specificity.

5.3 Dset_186-local Prediction

Table 4 reports strict prediction metrics on Dset_186-local. The holo-aware rows are retained only to show label-definition leakage and to analyze the optional

Table 4. External prediction on Dset_186-local (158 complexes, 39,505 residues).

Role	Model	Prec.	Rec.	F1 AUROC / AUPRC
Strict primary	EGNN	0.3605	0.3456	0.3529 0.7277 / 0.3050
Diagnostic	Holo-aware EGNN	0.6690	0.6223	0.6448 0.8868 / 0.6770
Diagnostic	Holo-aware cross-chain EGNN	0.6167	0.6735	0.6439 0.8891 / 0.5766

Table 5. External prediction on PDBtest_315-local (314 complexes, 65,119 residues).

Role	Model	Prec.	Rec.	F1 AUROC / AUPRC
Strict primary	EGNN	0.3057	0.3023	0.3040 0.6799 / 0.2675
Diagnostic	Holo-aware EGNN	0.7360	0.6252	0.6761 0.8946 / 0.7408
Diagnostic	Holo-aware cross-chain EGNN	0.7014	0.6629	0.6816 0.8983 / 0.6906

cross-chain module. The strict drop exposes a substantial generalization gap that must be addressed before claiming competitive benchmark performance.

5.4 PDBtest_315-local Prediction

Table 5 reports the completed PDBtest_315-local run. All 315 official chain-aware entries obtained structures from RCSB; 314 passed the current chain-quality rules. ESM-2 650M embeddings and both model predictions were produced on one NVIDIA GeForce RTX 5060 Laptop GPU. The strict model is reported first. Holo-aware EGNN and cross-chain rows are leakage diagnostics; within that diagnostic setting, cross-chain attention trades precision and AUPRC for recall and F1.

6 Discussion

The experiments show that a physically grounded label source and a language-model representation can form a reproducible engineering pipeline, but they also expose a critical evaluation hazard. Holo SASA is too close to the Δ SASA label definition to be accepted as a deployable input. ESM-2 contributes sequence and evolutionary context; structural descriptors and coordinates let the EGNN model local geometry. The true-650M correction is important: the final strict files contain 1,280 ESM dimensions and 1,287 total features, whereas older 320-dimensional embeddings would correspond to a different ESM-2 scale and cannot be compared as 650M experiments.

Cross-chain attention is therefore treated as an analysis module, not a stable core improvement. In the holo-aware diagnostic it improves recall and F1 on PDBtest_315-local, but does not consistently improve precision or AUPRC. A future strict training setup should explicitly sample non-interacting partner candidates.

7 Limitations and Reproducibility

This manuscript reports only experiments completed in the packaged project state. PDBtest_315-local preprocessing, ESM-2 extraction, and prediction are complete. The collector can be configured for 1,000+ complexes and the extractor supports larger ESM-2 models such as 3B, but those scale-up experiments should be run and reported separately with storage, runtime, and hardware details.

GraphPPIS is discussed as a representative prior method, but its published values are not inserted into our numeric tables because the current local preprocessing and split protocols are not guaranteed to match its official evaluation protocol. A direct comparison requires running both methods on the same manifests and labels.

All reported numeric tables are backed by CSV files included with this submission package under `data/`. The implementation exports per-residue predictions for benchmark analysis and stores model checkpoints with feature columns, normalization statistics, cutoff, hidden size, and metric metadata.

8 Conclusion

SASA-ESM provides an end-to-end implementation for weakly supervised PPI site prediction from protein complex structures. Under strict no-SASA inputs, true ESM-2 650M features and EGNN geometry reach internal $F1 = 0.7745$. External results are substantially lower and should be read as a reproducible baseline rather than a state-of-the-art claim. The leakage audit, exported predictions, and explicit diagnostic rows make the current package a more defensible foundation for future work.

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